

Well-structured Pandas questions (from basic → advanced → real-life)

- 1. What are Pandas?**
- 2. What are Series and DataFrame?**
- 3. How to import Pandas?**
- 4. How to read a CSV file?**
- 5. What is a DataFrame?**
- 6. How to display the first 5 rows?**
- 7. How to display the last 5 rows?**
- 8. How to check the shape of DataFrame?**
- 9. How to get column names?**
- 10. Difference between head() and tail()**
- 11. How to select a single column?**
- 12. How to select multiple columns?**
- 13. What is iloc?**
- 14. What is loc?**
- 15. Difference between loc and iloc**
- 16. How to select the first row?**
- 17. How to select specific rows and columns?**
- 18. How to filter rows based on condition?**
- 19. Filter students with marks > 50**
- 20. Filter data using multiple conditions**
- 21. Difference between & and and**
- 22. How to filter using query()**
- 23. How to filter null values**
- 24. What is Boolean indexing?**
- 25. Filter top 5 records**
- 26. What is missing data?**

- 27. What does dropna() do?**
- 28. Difference between dropna() and fillna()**
- 29. How to fill missing values with 0?**
- 30. How to fill with mean?**
- 31. What is drop_duplicates()?**
- 32. How to check null values?**
- 33. How to find mean of a column?**
- 34. How to find max and min values?**
- 35. What is value_counts()?**
- 36. What is unique()?**
- 37. Difference between unique() and nunique()**
- 38. How to use groupby()**
- 39. How to sort values?**
- 40. Difference between ascending and descending**
- 41. How to add a new column?**
- 42. How to update values using loc?**
- 43. What is rename()?**
- 44. How to drop a column?**
- 45. What is nlargest()?**
- 46. How to merge two DataFrames?**
- 47. Difference between merge and concat**
- 48. How to save DataFrame to CSV?**
- 49. How to take sample data (sample())?**
- 50. Explain real-life use of Pandas**
- 51. Find top 10 students by marks**
- 52. Find average marks**
- 53. Count students per city**
- 54. Find highest marks in each course**
- 55. Replace marks < 40 with 40**
- 56. Remove duplicate students**
- 57. Find students with attendance > 75**

58. Sort students by marks

59. Find number of unique courses

60. Create new column “Total Marks + Bonus”